

Response to the Reviewers

Manuscript ID: [ACCESS-2026-XXXXX] — “Excitation-Coherent Telemetry Analysis for Predictive OBD-II Connector Wear Detection: A Physics-Informed Study on Real In-Vehicle Network Data”

We thank the Editor and the Reviewers for their careful reading and constructive comments. Below, each comment is reproduced in bold, followed by our response and the corresponding changes. Changes in the revised manuscript are highlighted in blue.

Reviewer 1

Comment R1.1 (realism of the faults). *The faults are simulated; the detector may only learn the simulator.*

Response. We agree that this is the central question and address it in four ways. (i) All healthy telemetry, the excitation that drives the faults and the usage histories are real; only the connector fault is modelled, and the detector never sees model internals. (ii) Sixteen parameter perturbations (Fig. 9): on full-bus data the AUROC stays within 0.84–0.95 and severe-stage AUROC ≥ 0.97 . (iii) A *structurally different* fault process (bursty Markov chatter with an energy-based brown-out criterion; Supplement S8): trained on the nominal model, ECTA reaches 0.856 on full-bus data (severe 0.999). (iv) The analytical observability law (Section VI-C), which involves no classifier, predicts the measured detectability (Spearman 0.93).

Changes in the manuscript: [Section/page references.]

Comment R1.2 (stronger deep baselines). *Compare with a Transformer.*

Response. A Transformer encoder baseline is reported in Supplement S10 for all four corpora; it does not close the gap to ECTA on unseen vehicles.

Comment R1.3 (statistical significance).

Response. Paired vehicle-level bootstrap confidence intervals and p-values for ECTA against every baseline are given in Supplement S11.

Comment R1.4 (hyperparameters).

Response. Hyperparameters were fixed before evaluation and are identical across corpora. Supplement S9 shows that seven gradient-boosting configurations change the AUROC by at most a few thousandths.

Reviewer 2

Comment R2.1 (practical false-alarm rate).

Response. Supplement S12 reports false alarms per hour of real healthy driving at fixed detection rates. A full-bus tool detecting 90% of severely worn connectors raises no false alarm in the held-out healthy data; low-rate polling tools need multi-trip pooling (Fig. 10).

Comment R2.2 (window length).

Response. Supplement S13 reports 10-, 30- and 60-s windows; the 30-s choice is not critical.

Comment R2.3 (validation on naturally worn connectors).

Response. We agree that this is the most valuable next step and state it as the primary limitation. [If bench data have been collected: describe specimens, protocol and results here.]

Reviewer 3

Comment R3.1 [comment].

Response. [response]

Changes in the manuscript: [changes]